

Nilpotent Matrices

Topic.

Matrix algebra, nilpotent matrices, invertibility.

Statement.

A square matrix A is called nilpotent if

$$\exists n \in \mathbb{N} : \quad A^n = 0.$$

- (a) Let A be nilpotent. Prove that $E - A$ is invertible.
- (b) Prove that if A and B are nilpotent matrices of the same size and

$$AB = BA,$$

then $A + B$ is nilpotent.

Give an example of two nilpotent 2×2 matrices such that $A + B$ is not nilpotent.

Solution.

(a)

Let A be nilpotent. Then

$$\exists n \in \mathbb{N} : \quad A^n = 0.$$

Consider the matrix

$$S = E + A + A^2 + \cdots + A^{n-1}.$$

Then

$$(E - A)S = ES - AS.$$

Hence

$$(E - A)S = (E + A + \cdots + A^{n-1}) - (A + A^2 + \cdots + A^n).$$

After cancelling equal terms,

$$(E - A)S = E - A^n.$$

Since

$$A^n = 0,$$

we get

$$(E - A)S = E.$$

Similarly,

$$S(E - A) = E.$$

Therefore,

$$S = (E - A)^{-1}.$$

Thus, $E - A$ is invertible.

(b)

Let

$$A^p = 0, \quad B^q = 0$$

for some

$$p, q \in \mathbb{N}.$$

Consider

$$(A + B)^{p+q-1}.$$

Let

$$m = p + q - 1.$$

The expression $(A + B)^m$ is a product of m brackets

$$(A + B)(A + B) \cdots (A + B).$$

After expanding the brackets, each term is a product of length m consisting of matrices A and B .

Since

$$AB = BA,$$

any neighboring A and B can be interchanged. Therefore, in every product all matrices A can be moved to the left and all matrices B to the right.

Thus every term can be written in the form

$$A^k B^{m-k}, \quad 0 \leq k \leq m.$$

Since

$$m = p + q - 1,$$

we get

$$A^k B^{p+q-1-k}.$$

Take any such term.

If

$$k \geq p,$$

then

$$A^k = 0.$$

Hence the whole product is zero.

If

$$k \leq p - 1,$$

then

$$p + q - 1 - k \geq p + q - 1 - (p - 1) = q.$$

Therefore,

$$B^{p+q-1-k} = 0.$$

Again, the whole product is zero.

Thus every term in the expansion is zero, and hence

$$(A + B)^{p+q-1} = 0.$$

Therefore,

$$A + B$$

is nilpotent.

Example.

Let

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then

$$A^2 = 0, \quad B^2 = 0.$$

So both matrices are nilpotent.

But

$$A + B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$(A + B)^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = E \neq 0.$$

Therefore, $A + B$ is not nilpotent.

Indeed,

$$(A + B)^n \neq 0, \quad \forall n \in \mathbb{N}.$$

This example shows why the condition

$$AB = BA$$

is needed.